

Linear Regression and Classification

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Basic setup for Classification/Regression

TRAINING

 ML component

*Inputs, Examples,
Trials, Instances (labelled)*

Day of week
Temperature
Holiday
Price
Units sold

Training/
Learning/
Model Building/

Model

Regressor/
Classifier
Function/
Predictor/
Model

TESTING

Trained Model

read model

Testing

Test examples
(unlabelled)

Day of week
Temperature
Holiday
Price

Predictions

Evaluation

Metric:

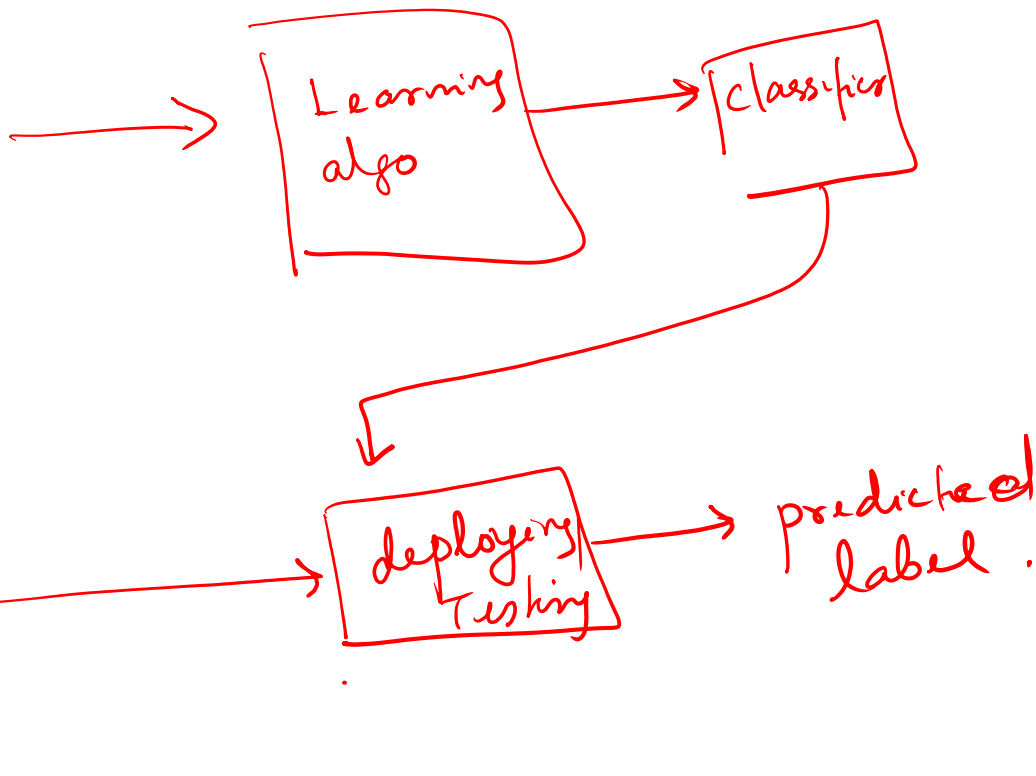
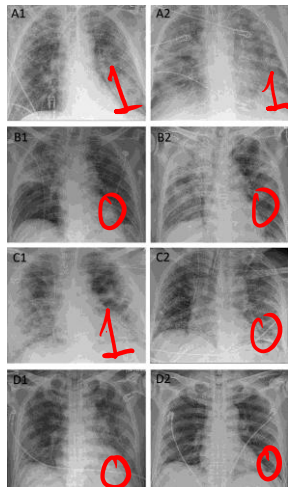
Mean square error, Mean absolute error for regression

0/1 error, cross-entropy, confusion matrix for classification

True values

Another example

Training data



Notations

An instance and its label



fixed dimensional vector
 x , $d = \#$ of dimension
5- features

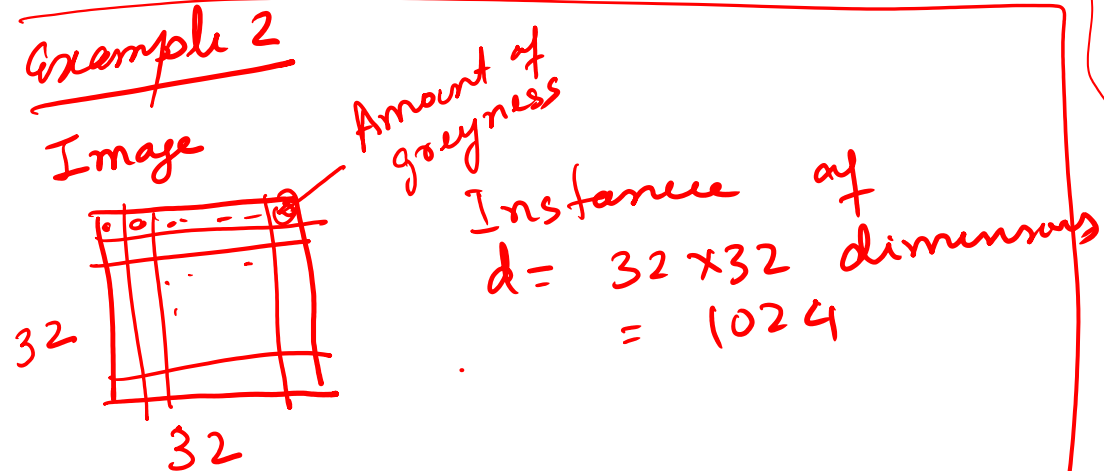
x_1 : CPI (Real 0-10)

x_2 : Minor in statistics?
0/1

x_3 : Honors or not?
0/1

x_4 : # of ML courses.
 $\{1, \dots, 5\}$

x_5 : # of BTP/R&D credits.



$y \equiv$ label of an instance

The training dataset: set of instances

N $\equiv$ # of training instances

N is typically large — can go into millions.

$$D = \{ (x^1, y^1), (x^2, y^2), \dots, (x^N, y^N) \}$$

set notation for training data

Matrix notation for D

$$X = \begin{bmatrix} x_1^1 & \dots & x_d^1 \\ \vdots & & \vdots \\ x_1^N & \dots & x_d^N \end{bmatrix}$$

$$Y = \begin{bmatrix} y^1 \\ \vdots \\ y^N \end{bmatrix}$$

The model and its parameters

$$f: x \rightarrow y$$

$$f(x) = \hat{y} \leftarrow \text{predicted label}$$

- * Model or function can take many different forms.
- * A crucial step in ML is selecting the correct function given the data type & available training data
- * Simple choice of $f(x)$: linear combination of co-ordinates of input.

Let us practice to build a model manually.

$d=1, N=3$
D: x_i y
 x^1 : 2 14
 x^2 : 1 8
 x^3 : -3 -16

Model
building:
You all →

$$f(x) = 6x_1 + 2$$

parameters of the
function.

$$f(x; \underline{w}) = \underline{w}_1 x_1 + \underline{w}_0$$

Test instance

$x: [0]$

$$\hat{y} = f([0]) = 6 \cdot 0 + 2 = 2$$

$d=2$

		y
x^1	$[1, 3]$	7
x^2	$[-1, 0]$	0
x^3	$[2, 1]$	7

Model class:

$$f(x; w) = \underline{w}_1 x_1 + \underline{w}_2 x_2 + \underline{w}_0$$

You
to train

$$f(x) : 2x_1 + x_2 + 2$$

But real data will not have perfect labels...

$[1, 3]$ ^y 6.1

$[-1, 0]$ 1

$[2, 1]$ 5.5

$[11, 3]$ 7.5

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$$f(x; w) = w_1 x_1 + w_2 x_2 + w_0$$

Challenges to be addressed

- Space of possible functions extremely large: need to limit the set from which f is chosen (Hypothesis family)
 - E.g. polynomials
 - Piecewise constant boxes
 - Neural networks...
- How to ensure f 's performance on D will generalize to unseen instances
- How to fit the function on the training dataset tractably.